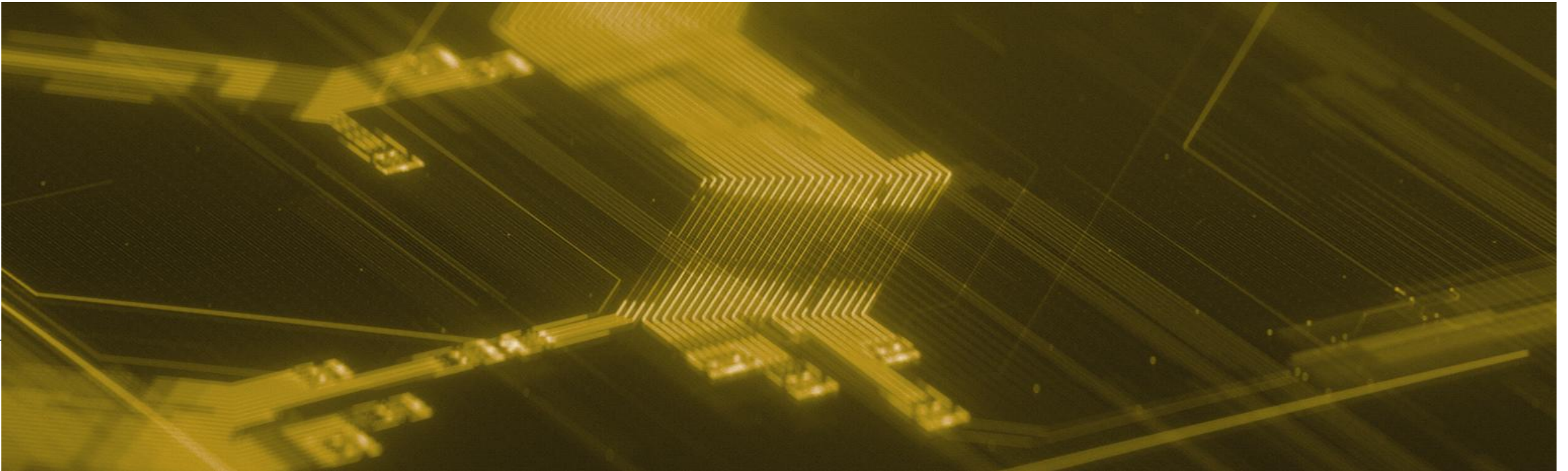


Predicting Aviation Accident Severity

A Dual-Branch Machine Learning Architecture
for Imbalanced Safety Data



One Team



Hanyu Jee

Handong Global University
Data Science & Statistics

Team lead



Minsol Kim

Jeonbuk Nat'l Univ
Statistic & Economics

Tabular Branch
Modeling



Junho Lee

Seoul Nat'l Univ
BME & Business

Preprocessing & EDA
Optimization



Dana Jeong

Gyeongsang Nat'l Univ
Electronic Engineering

Text Branch Modeling



Jaewon Kim

Gyeonggi University of
Science Technology

Presentation Design &
Documentation

Overview

01 Recent Trends in Aircraft Accidents
Where fatal risk still concentrates

02 Research Objectives & Data
NTSB dataset and feature selection

03 Overall Analysis Pipeline
The dual-branch architecture

04 Text Branch — SafeAeroBERT + MLP
Learning from investigator narratives

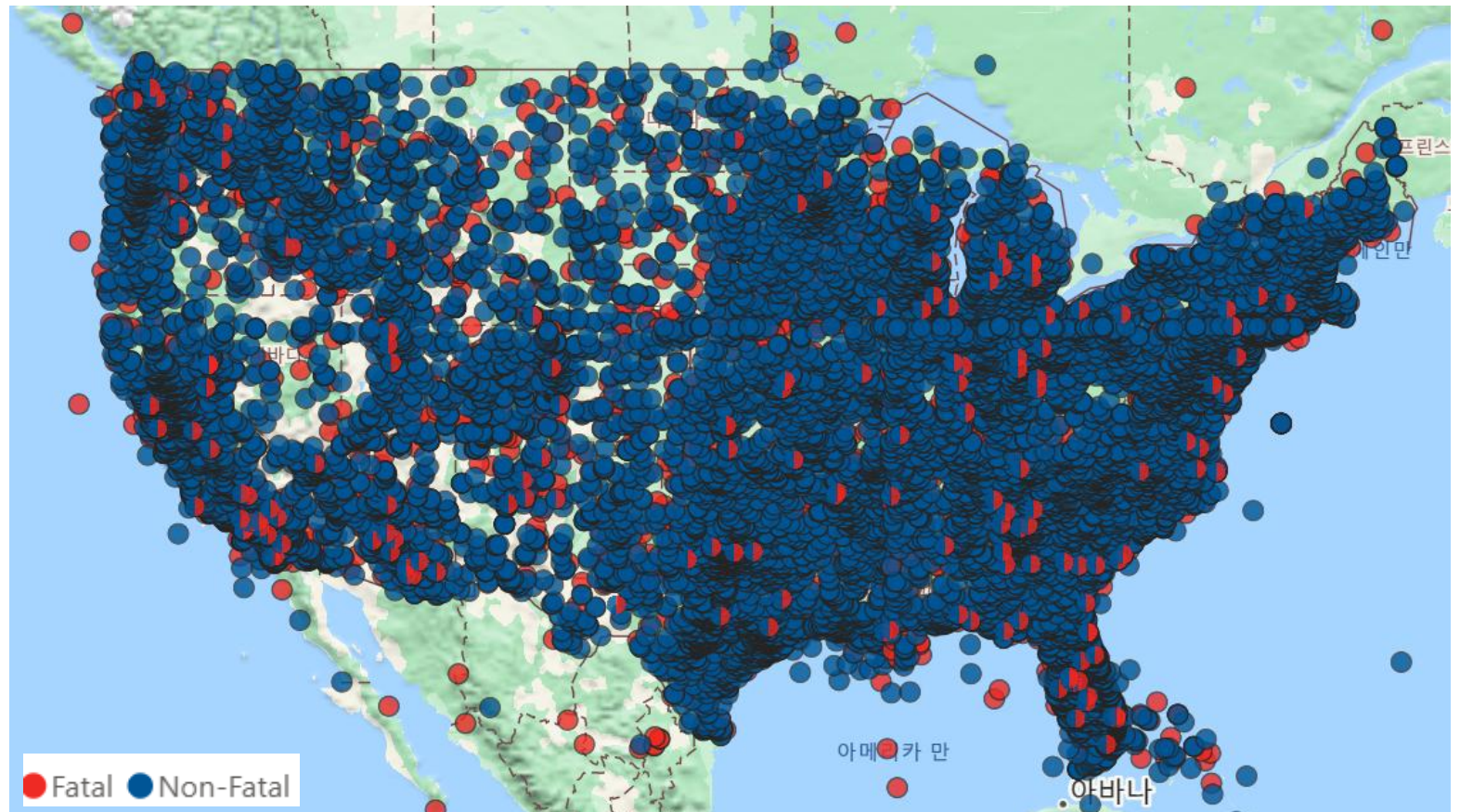
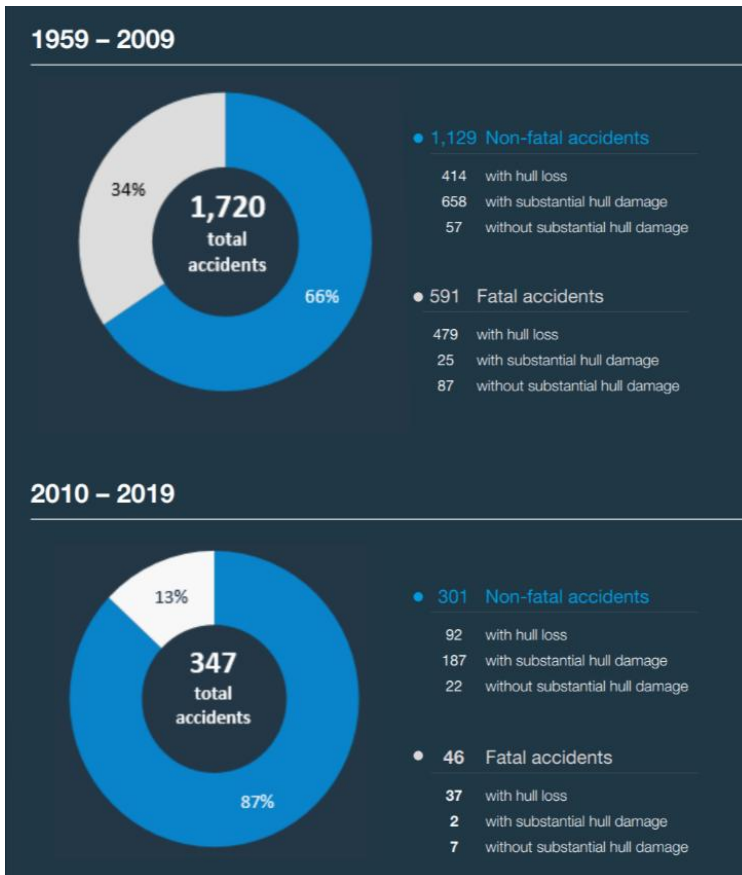
05 Tabular Branch — Classification
Tree ensembles on structured data

06 Conclusion & Appendix
Fusion plan and supporting analysis

Recent Trends in Aircraft Accidents

Across the global commercial jet fleet, total accidents fell dramatically between the two periods, yet fatal events remain a persistent share of all occurrences — the core motivation for a severity-focused model.

(Boeing & NTSB statistical reports)



Research Purpose

01 Automated Accident Severity Classification

The aviation industry generates massive volumes of unstructured accident reports. Efficiently processing this big data is critical for the optimal allocation of limited investigation resources.

02 Data-Driven Safety Management

Build a machine-learning architecture that automatically classifies the severity of post-event occurrences from factual data and investigator narratives.

03 Paradigm Shift: Classification → Statistics → Policy

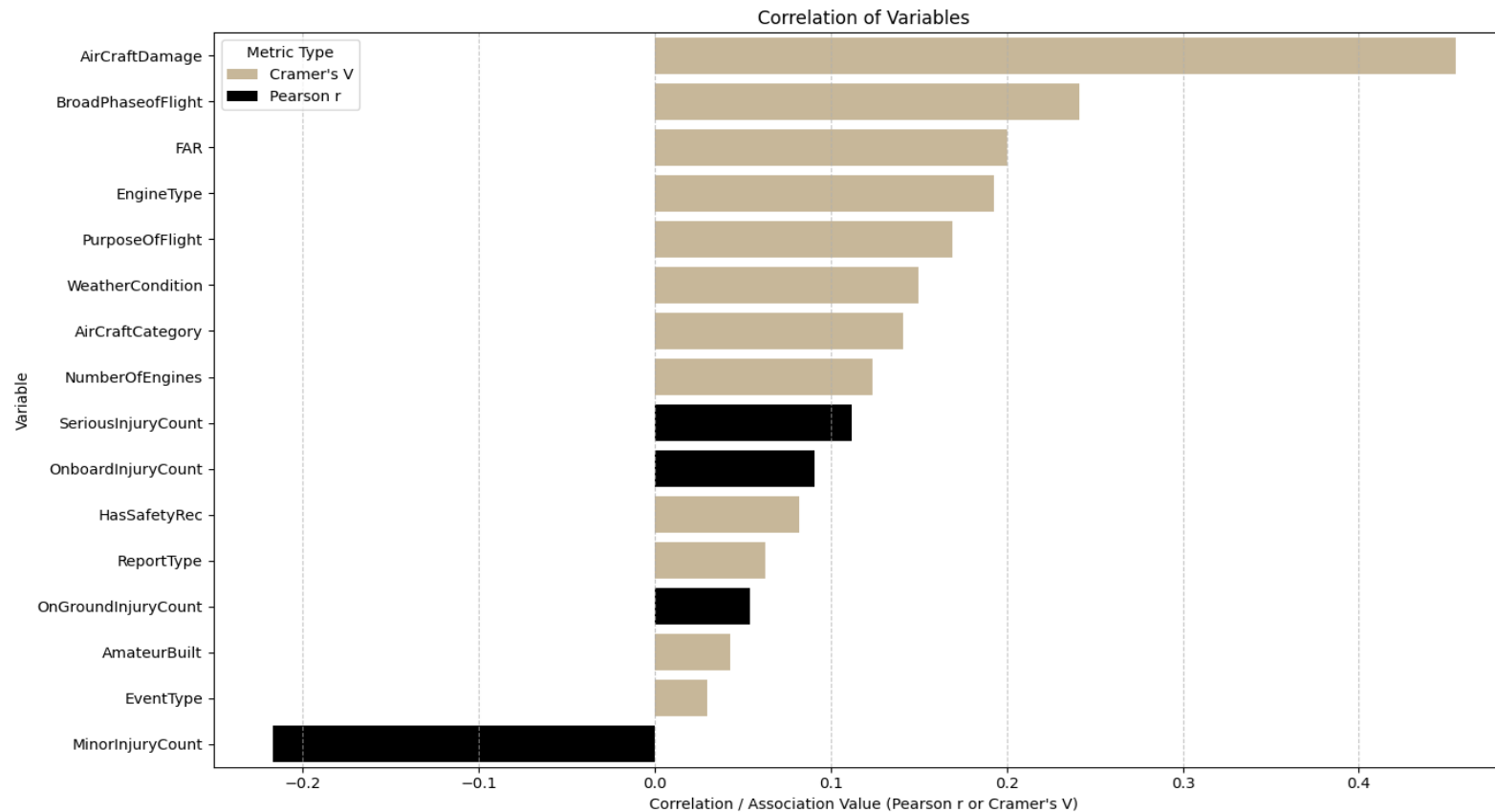
Beyond simple categorization, accumulating classified data unlocks macro-level risk-pattern analysis to inform safety policy.



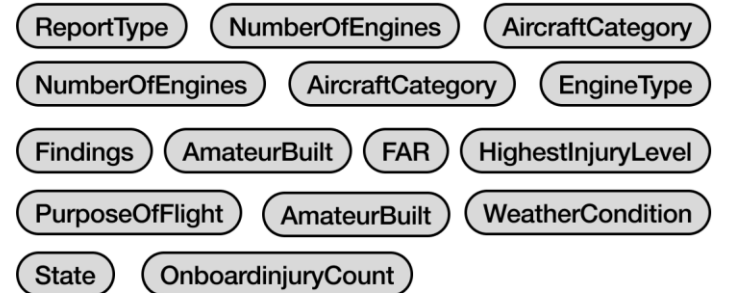
Feature Selection

Input variables were selected by comparing their association with the Highest Injury Level.

Variables with relatively stronger associations were prioritized while considering their availability and practical usefulness for prediction.



Selected Input Variables



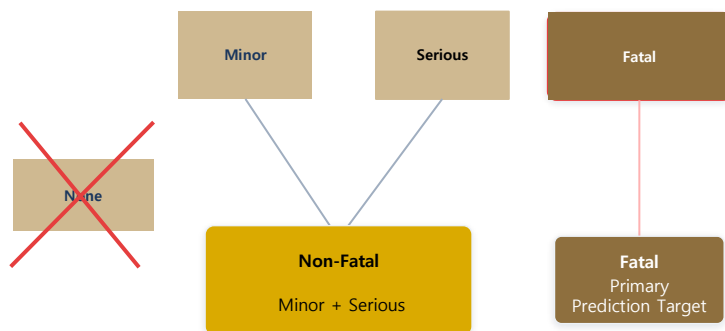
Dataset Description

The NTSB dataset contains detailed investigation records of transportation accidents and incidents — aviation, railroad, highway, and marine.

- | | |
|--|--|
| 01 HighestInjuryLevel — extent of injury caused by the accident. | 07 AircraftCategory — e.g. Airplane, Helicopter. |
| 02 Findings — investigator narrative of contributing factors. | 09 EngineType — e.g. Jet, Turboprop, Piston. |
| 03 WeatherCondition — flight weather, e.g. VMC or IMC. | 11 FAR — applicable regulation, e.g. Part 121, Part 91. |
| 04 BroadPhaseofFlight — flight phase at the time of the accident. | 08 NumberOfEngines — engines installed on the aircraft. |
| 05 State — U.S. state where the accident occurred. | 10 AmateurBuilt — amateur-built aircraft (Yes / No). |
| 06 AircraftDamage — damage extent: Minor, Substantial, Destroyed. | 12 PurposeOfFlight — e.g. Personal, Training, Commercial. |

METHODOLOGY

Data Cleaning & Transformation



PARTITION	ALL	NON-FATAL	FATAL
Train (80%)	2,593	1,678	915
Validation (10%)	324	210	114
Test (10%)	324	209	115

Total 3,241 records · Non-Fatal 2,097 · Fatal 1,144

Minor + Serious → Non-Fatal 2,097 rows

Fatal → Fatal 1,144 rows

Fatal is the primary prediction target. A stratified 80 : 10 : 10 split (SEED = 123) preserves the class ratio in every partition.

TARGET

Binary reframing

Removed None (no-injury) records and collapsed the ambiguous 4-class label into **Fatal / Non-Fatal** — eliminating class confusion and, operationally, guarding against the most catastrophic error: missing a fatal accident.

SPLIT

Train / Val / Test

Stratified **80 : 10 : 10** split (SEED = 123) preserving the Fatal / Non-Fatal ratio across all partitions so evaluation reflects the true class balance.

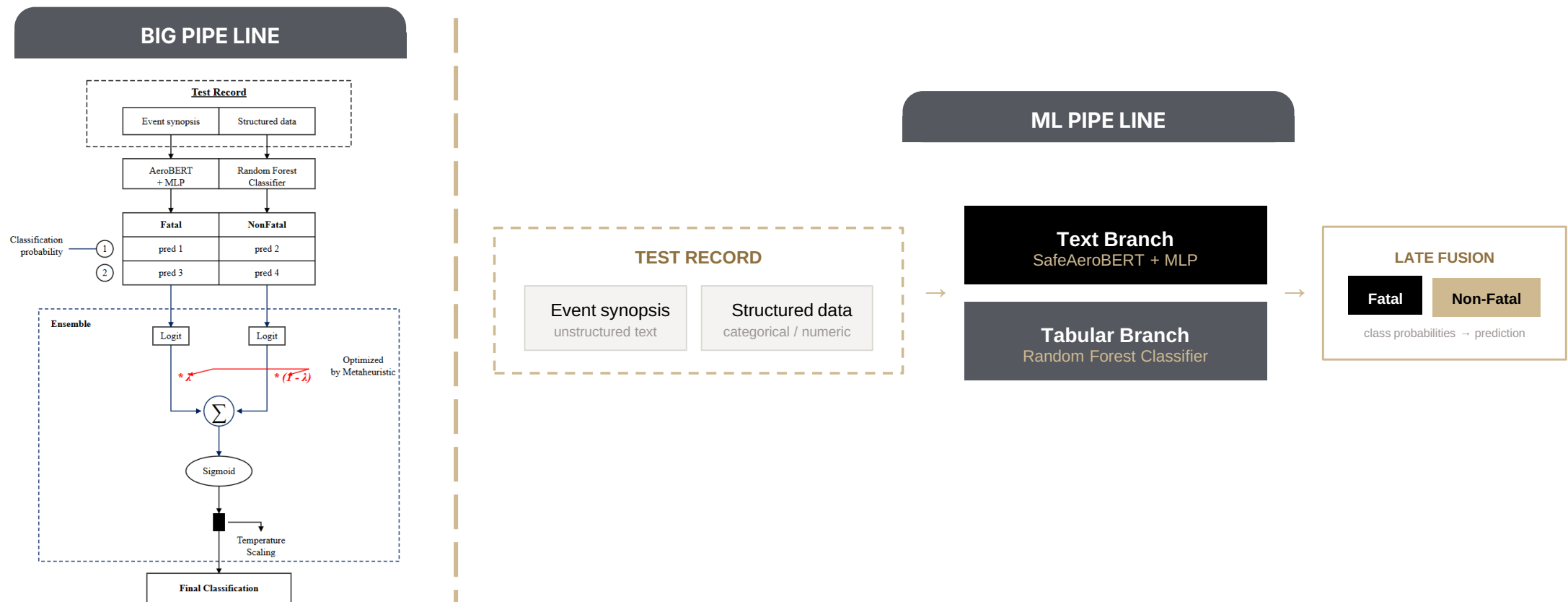
METRICS

Recall over accuracy

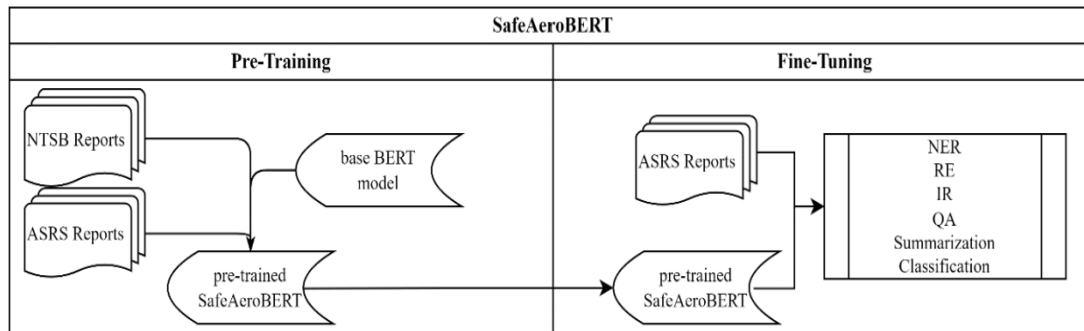
Although Accuracy is reported, Fatal Recall is prioritized because detecting fatal accidents is the primary objective. Macro-F1 is used to evaluate balanced performance across classes.

Research Design: Dual-Branch Pipeline

Prior work independently applied SVM to unstructured text and KNN to structured data, showing that separate learning approaches are necessary. We adopt that parallel architecture (Anushree & Sowmya, 2020).



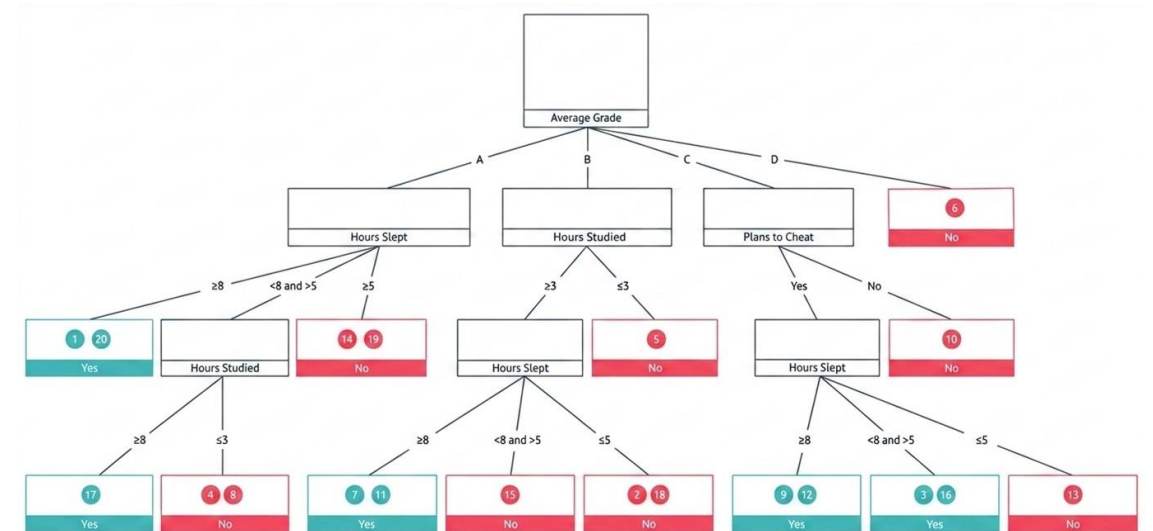
Method Selection



TEXT BRANCH

SafeAeroBERT

Rather than a general language model, we adopt a BERT embedding specialized for the aviation domain — capturing the safety-report terminology and investigation context that generic models miss.



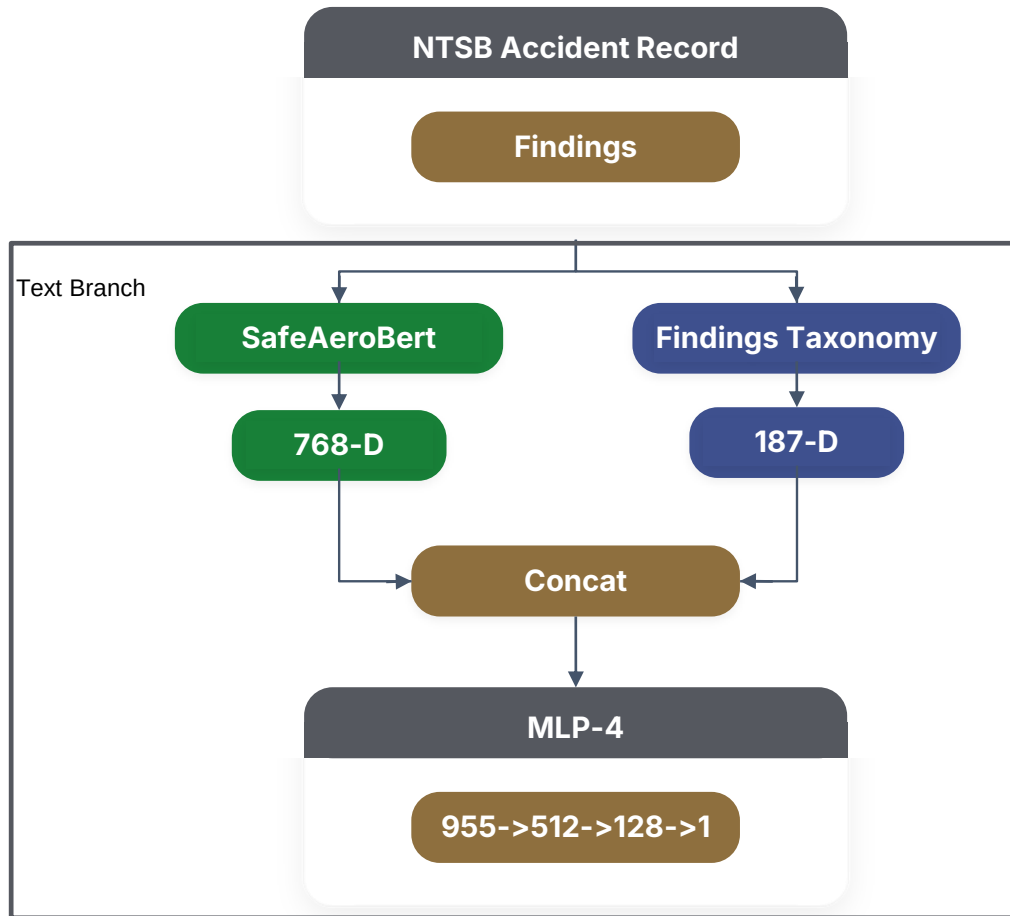
TABULAR BRANCH

Tree-based Ensemble

To capture the non-linear relationships among structured categorical features, we use tree-based ensembles instead of deep learning — a choice supported by prior aviation-safety research.

Feature Architecture & Findings Taxonomy

Text Branch Architecture



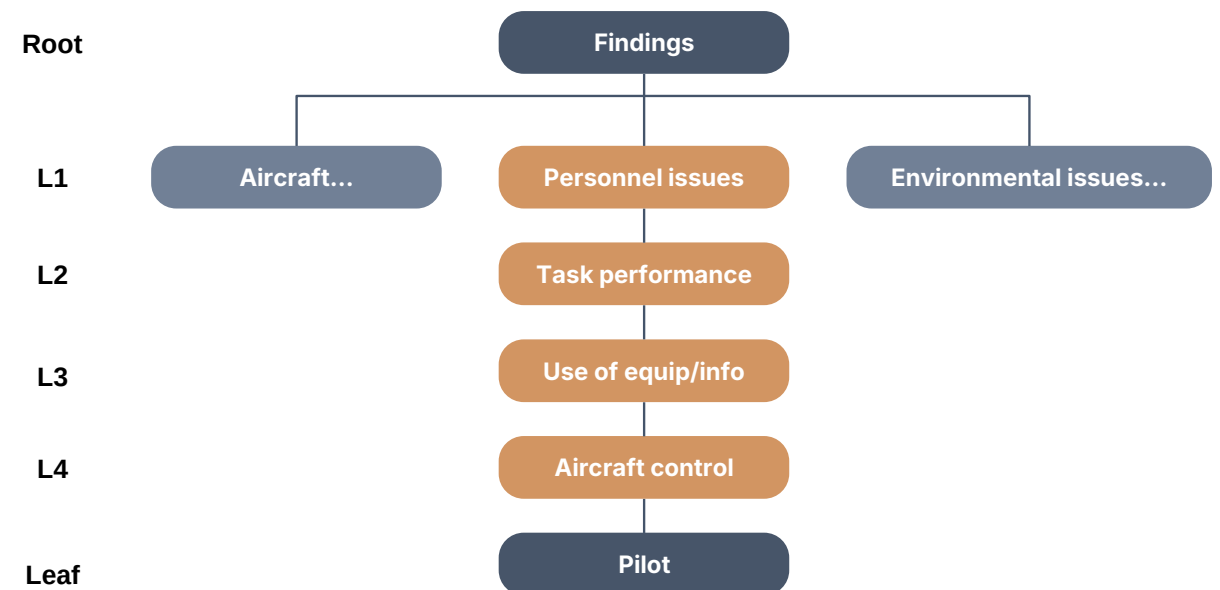
Why SafeAeroBERT?

Findings contain aviation-investigation-specific terminology and domain context. SafeAeroBERT captures the semantic information of safety reports that a general model would miss.

Why the Findings Taxonomy?

Findings record accident causes as hierarchical paths from L1 to leaf level. Taxonomy features let the model identify causal hierarchies and structural information.

Taxonomy Hierarchy



* 180 train-only hierarchy tokens + 7 structural statistics = 187-D

TEXT BRANCH

MLP Architecture Comparison

Fixed: Weighted BCE (pos_weight = 1.83) · AdamW (lr = 5e-4) · dropout 0.20 · batch size 64 — only depth & width varied.

VALIDATION SET

MLP	LAYERS	ACC	MACRO F1	FATAL RECALL
MLP-1	955 → 128 → 1	0.781	0.752	0.623
MLP-2	955 → 256 → 1	0.775	0.750	0.658
MLP-3	955 → 256 → 64 → 1	0.769	0.749	0.693
MLP-4	955 → 512 → 128 → 1	0.750	0.740	0.781

To reduce the failure of missing fatal accidents,
MLP-4 was selected — accepting a small Macro-F1 trade-off for the highest Fatal Recall.

MLP-4 ON TEST SET

	PRED NON-FATAL	PRED FATAL	
ACTUAL NON-FATAL	148	62	
ACTUAL FATAL	12	103	
0.772 ACCURACY	0.768 MACRO F1	0.736 FATAL F1	0.896 FATAL RECALL

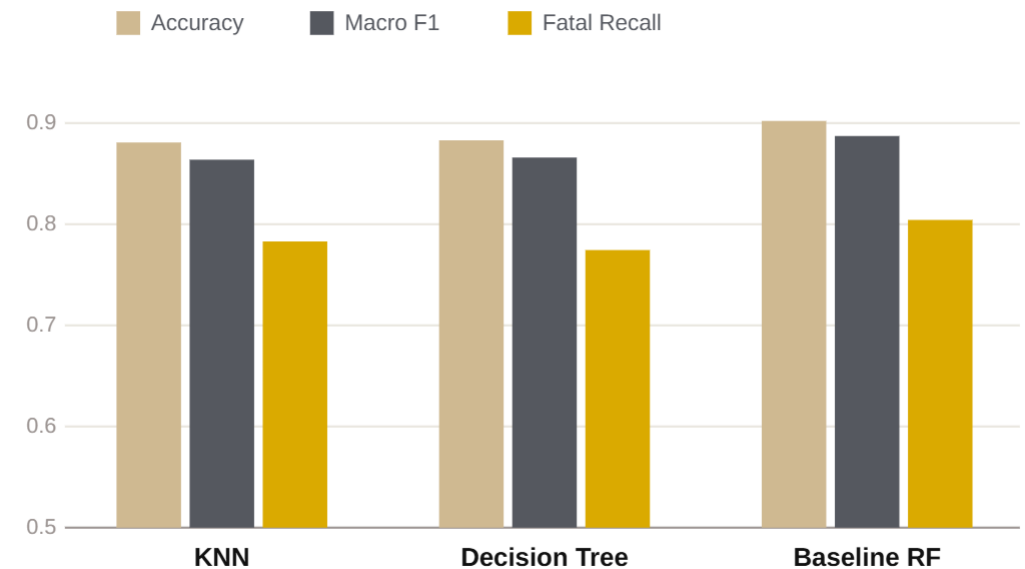
Initial Modeling & Performance Comparison

Model Selection Rationale

- Selected representative classifiers with distinct learning principles to provide a comprehensive baseline comparison.
- Compared distance-based (KNN), tree-based (Decision Tree), and ensemble-based (Random Forest) algorithms.

MODEL	CONFIG	ACC	MACRO F1	FATAL RECALL
KNN	k = 5	0.778	0.742	0.579
Decision Tree	random_state = 123	0.784	0.746	0.561
Baseline RF	n_estimators = 100 · balanced	0.821	0.790	0.623

CONCLUSION Baseline Random Forest selected



Handling Class Imbalance & Improving Random Forest

Balanced Random Forest

Applied to reduce class imbalance and improve **Fatal Recall**.

Tuned Random Forest (Class-weighted)

Applied hyperparameter tuning and class weighting to achieve a better balance between **Accuracy, Macro F1, and Fatal Recall**.

One-vs-Rest Random Forest

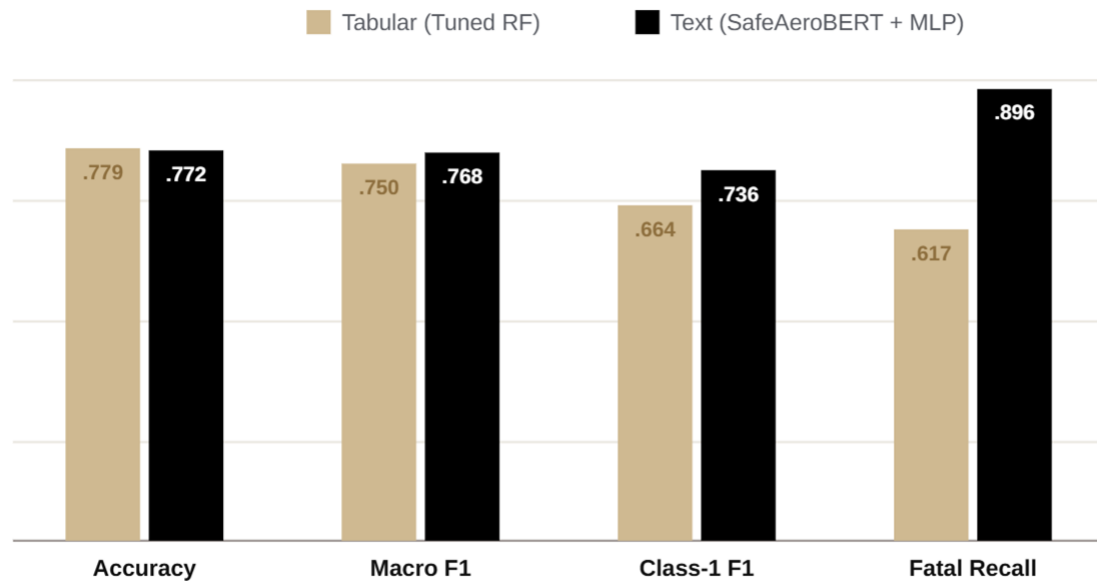
Applied to generate class probabilities for the **Late Fusion** framework.

MODEL	CONFIGURATION	ACCURACY	MACRO F1	FATAL RECALL	AUC
Balanced Random Forest	n_estimators=100 random_state=42	0.784	0.760	0.658	-
Tuned Random Forest (Class-weighted)	n_estimators=300 max_features=5 min_samples_split=5 criterion=gini class_weight=balanced	0.799	0.777	0.684	0.853
One-vs-Rest Random Forest	n_estimators=300 max_features=5 min_samples_split=5	0.809	0.771	0.579	0.857

Final Selection

Tuned Random Forest achieved the best overall balance, including the highest Fatal Recall (0.684), and was selected as the final **Tabular Branch model**.

Final Model Performance — Test Set



- 01 The **Text Branch** learns accident causes directly from Findings and achieves the highest **Fatal Recall**.
- 02 The **Tabular Branch** is steady on structured data but trails on Fatal Recall.
- 03 Because the branches are **complementary**, late fusion and optimization combine them to improve final prediction.

References

- D NTSB Accident Reports (dataset). Zenodo. <https://zenodo.org/records/17096333>
-
- 1 Siow, B. Y. (2025). *A Practical Approach to using Supervised Machine Learning Models to Classify Aviation Safety Occurrences*. arXiv:2504.09063v1.
-
- 2 Monika, Verma, S., Kumar, P., & Kakran, M. (2021). A Stacking Model for Aviation High-Risk Event Decision Making. *J. Physics: Conf. Series*, 1950(1), 012069.
-
- 3 Maynard, E., & Harris, D. (2022). Using neural networks to predict high-risk flight environments from accident and incident data. *Int. J. Occupational Safety and Ergonomics*, 28 (2), 1204–1212.
-
- 4 Anushree, H. R., & Sowmya, B. P. (2020). Ensembled machine learning model for aviation incident risk prediction. *IJRTE*, 9(3), 351–353.
-
- 5 Nanyonga, A., Joiner, K. F., Turhan, U., & Wild, G. (2025). Deep learning approaches for classifying aviation safety incidents: Evidence from Australian data. *AI*, 6(1), 251.
-

AI Utilization

- 0 Project topic and dataset candidate exploration — **Perplexity**
- 1 Prior research and related literature review — **Gemini Deep Research**
- 2 BERT+MLP and Random Forest code development — **Codex**
- 3 Data splitting, row alignment, data leakage checks, and code debugging — **Codex**
- 4 Evaluation metrics, result visualization, and model interpretation support — **Codex**
- 5 Initial presentation slide development — **Claude**

The researcher independently defined the research objectives, selected the dataset, designed the model architecture, and made all final decisions.

AI tools were used only to support research, code drafting, error checking, and documentation.

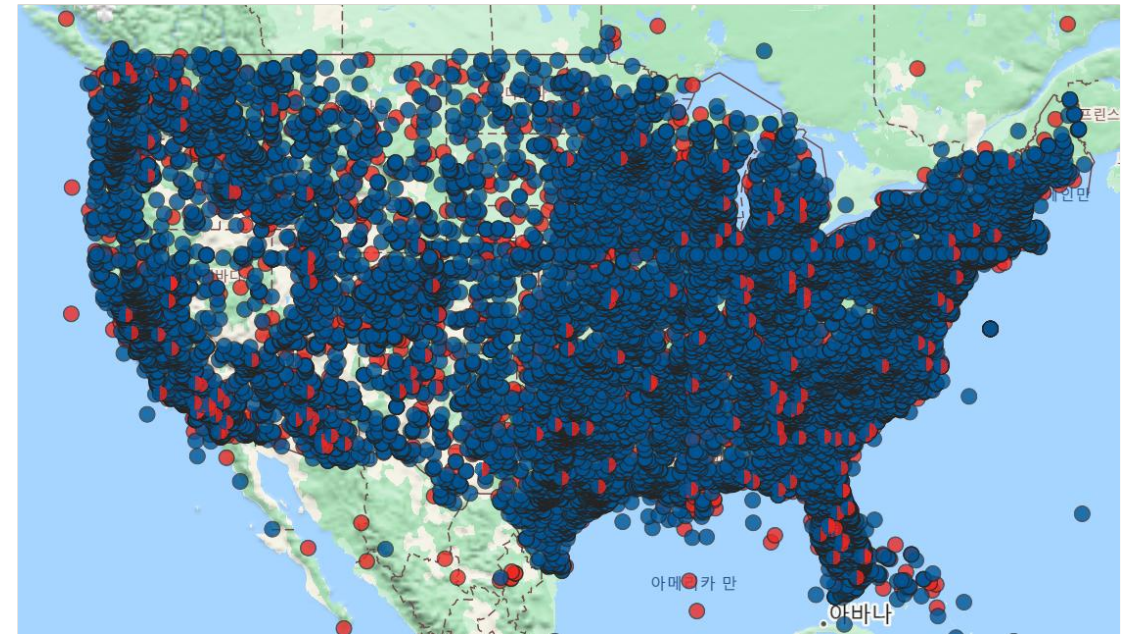
Thank You.

Appendix

Exploratory Data Analysis

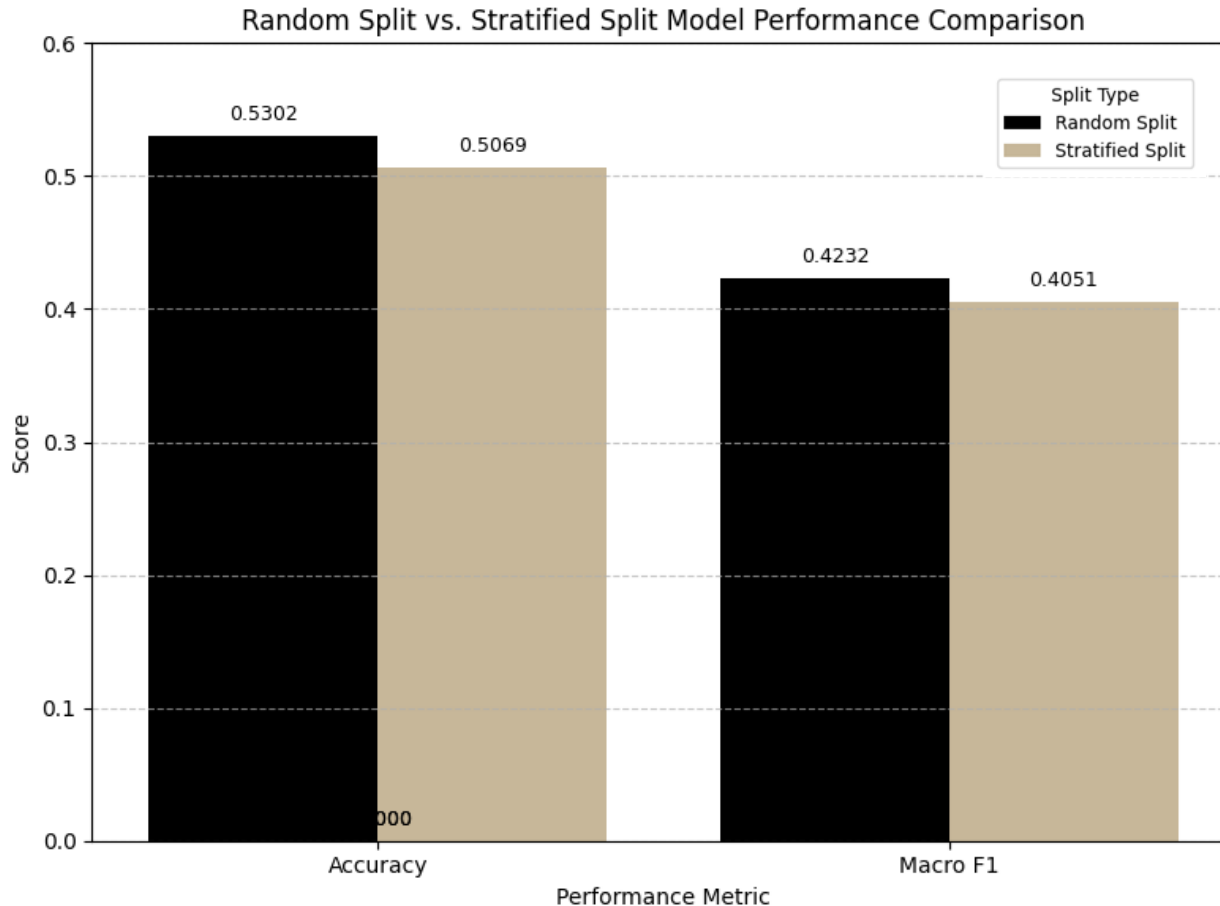
- 01 Severity was reframed as a binary **Fatal / Non-Fatal** target to sharpen fatal-accident detection.
- 02 Under **IMC** , the fatal-accident ratio is relatively higher — weather conditions relate to severity.
- 03 Fatal occurrence patterns differ by **flight phase** and by **purpose of flight**.

→ Selected: Findings, AmateurBuilt, NumberOfEngines, PurposeOfFlight, FAR, State, WeatherCondition, BroadPhaseofFlight, EngineType, AircraftCategory, HighestInjuryLevel, AircraftDamage.



Geographic distribution of accidents (blue = Non-Fatal, red = Fatal)

Data Splitting Strategy



Comparison

Accuracy and **Macro F1** showed only minor differences between **Random Split** and **Stratified Split**.

- **Stratified Split** maintained performance comparable to **Random Split**.

Bias–Variance Analysis

- **Average Bias:** 0.4215
- **Average Variance:** 0.1485
- The relatively low variance indicates that model performance is **stable across different data splits**.

Final Decision

- **Stratified Split** preserves the class distribution across the **Train, Validation, and Test** sets, reducing evaluation bias and variance.
- Therefore, **Stratified Split (8:1:1)** was adopted for all subsequent experiments.

Target Redefinition I : OverSampling

Initial Modeling (4-Class Classification)

Model	Validation accuracy	Validation macro F1	Class 1 (fatal) recall
KNN	0.7716	0.7309	0.5439
Decision tree	0.7747	0.7382	0.5702
Random forest	0.7840	0.7500	0.5702

- Compared **KNN**, **Decision Tree**, and **Random Forest**
- **Random Forest** achieved the best Accuracy (**0.784**) and Macro F1 (**0.750**)
- **Fatal Recall remained low (0.570)** due to severe class imbalance
- **Macro F1** and **Fatal Recall** were selected as key evaluation metrics

Solution) OverSampling

Applied Oversampling to reduce class imbalance

Tuned Random Forest using **Grid Search** ($mtry = 5$, $ntree = 300$)

Improved minority-class Recall and prediction balance

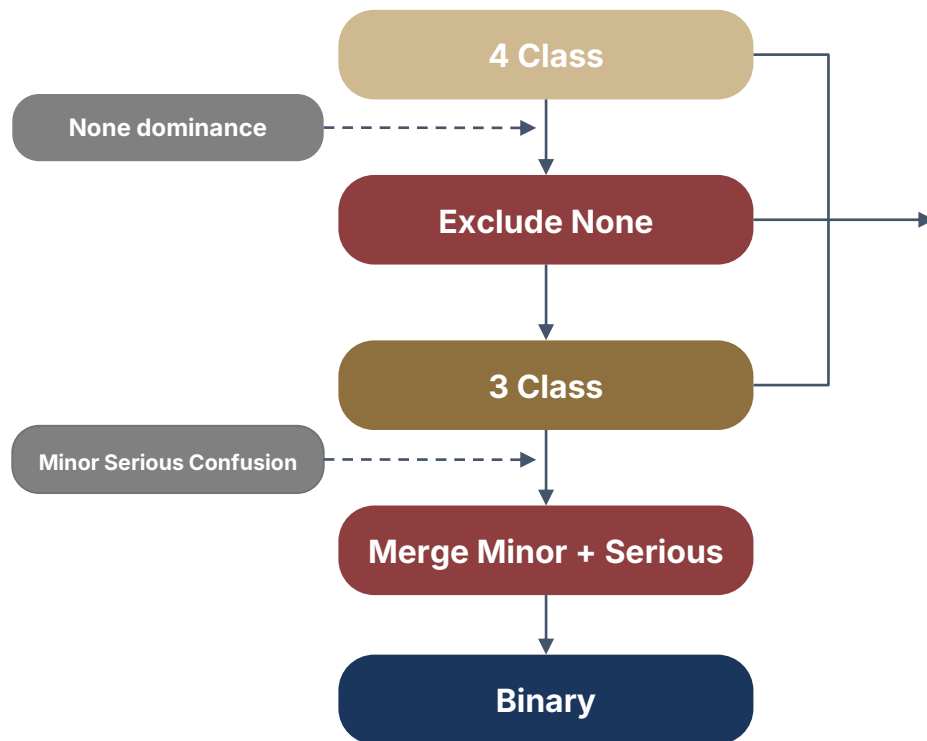
→ Remaining confusion between minority classes motivated **OvR** and **Binary Classification**

Target Redefinition II : 4-Class → 3-Class

The initial target had four classes (None, Minor, Serious, Fatal).

None accounted for 57.2% of the data, biasing the model toward injury *presence* rather than *severity*.

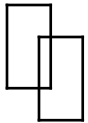
Excluding None sharpens the objective: predicting the severity of injury-producing accidents.



TARGET STAGE	None	Minor	Serious	Fatal	Total
Initial 4-class	4,336	1,189	908	1,144	7,577
ratio	57.2%	15.7%	12.0%	15.1%	100%
None excluded → 3-class	—	1,189	908	1,144	3,241
3-class ratio	—	36.7%	28.0%	35.3%	100%

Why exclude None

reduce the dominance of the no-injury class and distinguish severity *within* injury-producing accidents.



Target Redefinition III: 3-Class → Binary (in Text)

3-class experiments showed persistent **Minor–Serious confusion** :
raising one class's recall lowered the other's recall and Macro-F1. Merging Minor and Serious into **Non-Fatal** yields the final binary target.

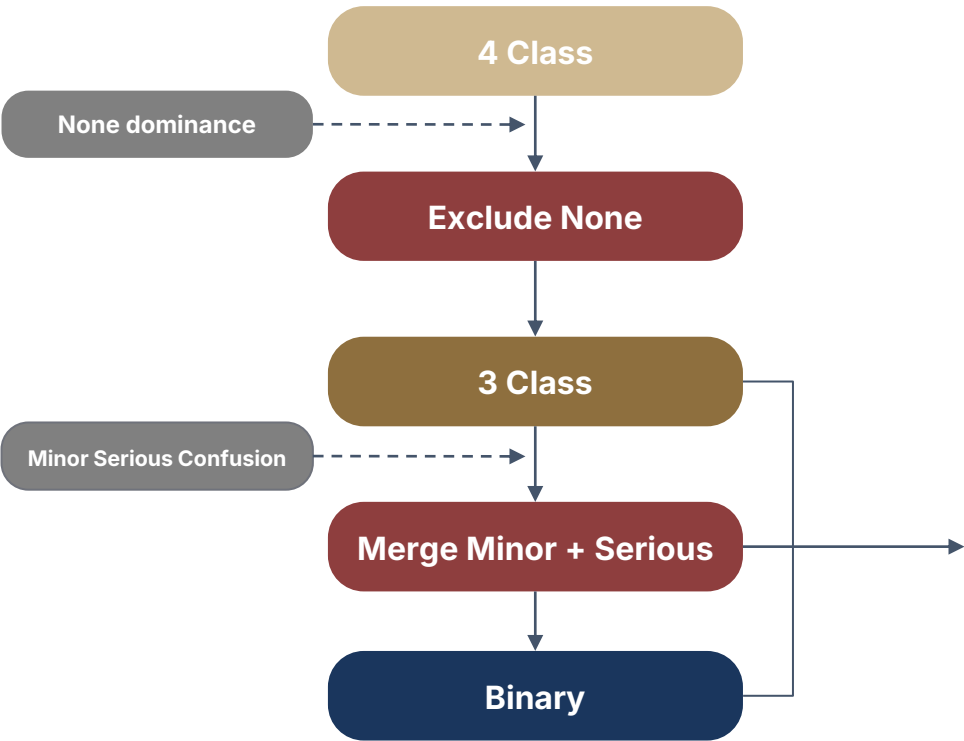


TABLE 1 · TEXT 3-CLASS CONFUSION MATRIX

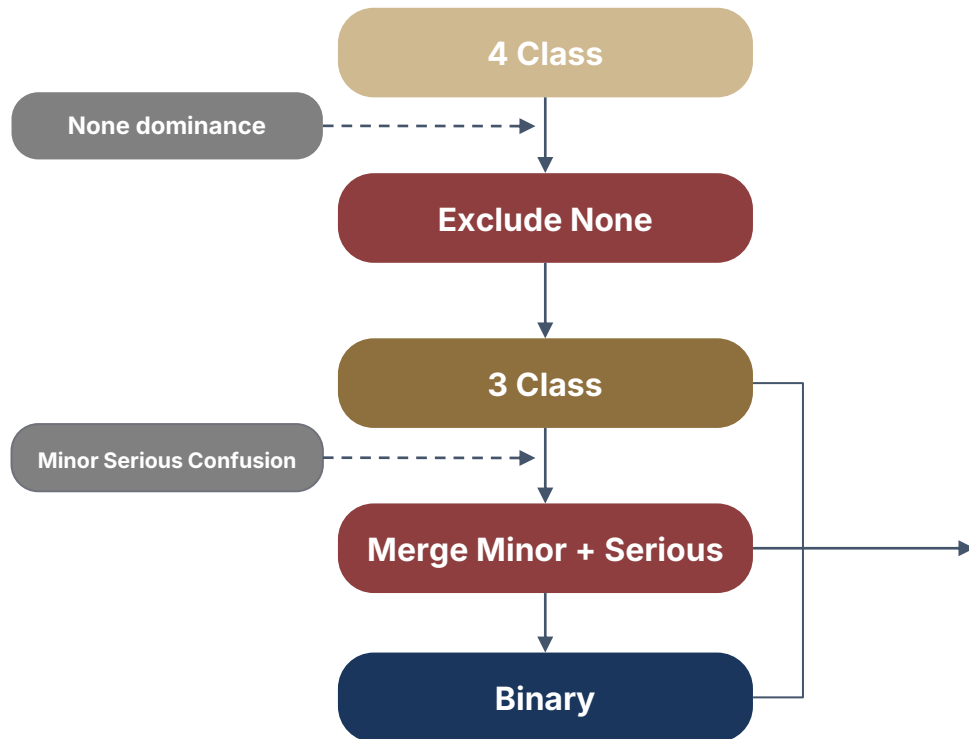
ACTUAL \ PRED	Minor	Serious	Fatal	Recall
Minor	59	35	24	50.0%
Serious	32	37	22	40.7%
Fatal	7	24	81	72.3%

TABLE 2 · MINOR–SERIOUS RECALL TRADE-OFF

SETTING	Minor R	Serious R	Fatal R	Macro F1
OVR Argmax	50.0%	37.4%	67.9%	52.0%
Serious weight ↑	13.6%	67.0%	79.5%	47.4%
Change	−36.4pp	+29.6pp	+11.6pp	−4.6pp

Boosting Serious recall collapses Minor recall and reduces Macro-F1
— the two classes cannot be separated stably, motivating the binary merge.

Target Redefinition III : 3-Class → Binary (in Tabular)



MODEL	TARGET	ACCURACY	MACRO F1	FATAL RECALL
4-Class RF	None / Minor / Serious / Fatal	0.784	0.750	0.570
3-Class RF	Minor / Serious / Fatal	0.600	0.570	0.630
Binary RF	Non-Fatal / Fatal	0.779	0.750	0.617

4-Class RF

High overall performance, but Fatal Recall stays low due to the dominance of the None class.

3-Class RF

Removing None improved Fatal Recall, but severe Minor–Serious confusion collapsed overall performance.

Binary RF

Merging Minor + Serious into Non-Fatal restored overall performance with more stable fatal-accident detection.

Model Improvement Process

MODEL	ACCURACY	MACRO F1	FATAL RECALL
BERT + MLP + Taxonomy	0.8123	0.8032	0.8435
Random Forest (text)	0.7785	0.7587	0.6957
HistGradientBoosting	0.7538	0.7490	0.8696
Logistic Regression	0.7723	0.7480	0.6522
Linear SVM	0.7138	0.6865	0.5913

CONCLUSION

- **SafeAeroBERT + Taxonomy + MLP-4** achieved the highest Accuracy and Macro-F1.
- HistGradientBoosting had slightly higher Fatal Recall but far weaker overall performance.
- **MLP-4** was selected for the best balance between fatal detection and general classification.